FACULTY OF ENGINEERING

Alexandria University

Department of Computer and Communications Engineering

Speech Recognition System

Research, Implementation & Evaluation Project

Phase 1: Research and Analysis Report

Team Members

Karim Mohamed — ID: 8237

Youssef Attia — ID: 8164

Mohamed Maher— ID: 8070

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Paper 1: ArTST — Arabic Text and Speech Transformer

Full Citation (IEEE): H. O. Toyin, A. Djanibekov, A. Kulkarni, and H. Aldarmaki, "ArTST: Arabic Text and Speech Transformer," in *Proceedings of ArabicNLP 2023 (co-located with EMNLP 2023),* Singapore, Dec. 7, 2023, pp. 41–51. DOI: 10.18653/v1/2023.arabicnlp-1.5

A. Summary and Context

Main Topic and Research Objectives

ArTST (Arabic Text and Speech Transformer) is a monolingual Arabic pre-trained model built from scratch on Arabic speech and text data, targeting three downstream tasks: Automatic Speech Recognition (ASR), Text-to-Speech synthesis (TTS), and spoken dialect identification. The central research objective is to demonstrate that language-targeted pre-training — designing and training a model specifically for Arabic rather than adapting an English-centric or multilingual foundation — leads to superior performance with far fewer parameters and computational resources. The paper challenges the prevailing assumption in the NLP and speech communities that large multilingual models, trained on hundreds of thousands of hours of multilingual data, are the optimal solution for any target language.

Specific Problem Addressed in ASR

The paper specifically addresses Modern Standard Arabic (MSA) ASR, confronting two linguistically unique challenges:

- Diacritization in ASR: Written Arabic typically omits short vowels (diacritics), creating an inherent ambiguity where the same written word can map to multiple pronunciations and meanings depending on context. Standard ASR systems trained on undiacritized text transcriptions must therefore learn to infer pronunciation from context alone. The paper uniquely explores whether ASR itself can serve as a diacritization tool by training on diacritized targets — a novel research direction with practical applications in Arabic education technology and text normalization.
- Morphological richness: Arabic's root-and-pattern morphological system means that a single root can yield hundreds of surface word forms, producing extremely large vocabulary sizes and high out-of-vocabulary rates. This severely challenges language model integration and degrades recognition performance compared to morphologically simpler languages like English.
- MSA/Dialect mixing in broadcast data: The MGB-2 corpus, which serves as the primary training resource, contains approximately 30% dialectal Arabic content despite being nominally MSA, creating training instability and evaluation ambiguity.

Key Contributions and Innovations

- First fully open-source Arabic-native SpeechT5-based transformer, pre-trained entirely from scratch on approximately 1,000 hours of Arabic speech — establishing a new benchmark for Arabic speech technology accessibility.
- Demonstrated that monolingual Arabic pre-training produces ASR performance (12.8% WER on MGB-2) that is at least twice as accurate as large multilingual models including Whisper medium, Whisper large, and MMS, while requiring a fraction of the parameters.

- Established a new state-of-the-art on the ADI17 Arabic spoken dialect identification benchmark, achieving 94% classification accuracy across 17 Arabic dialects — a notable result given the model was pre-trained exclusively on MSA data.
- Introduced a unified encoder-decoder architecture that handles speech and text as both input and output modalities within a single pre-trained model, enabling ASR, TTS, and dialect identification with the same backbone through task-specific fine-tuning heads.
- Received the Best Paper Award at ArabicNLP 2023 and released all pre-trained and fine-tuned model weights publicly on HuggingFace, making reproduction and extension accessible to the research community.

Significance and Potential Impact on Arabic Speech Technology

ArTST represents a paradigm shift in Arabic speech research: rather than defaulting to large-scale multilingual pre-training, it rigorously demonstrates that language-specific models can outperform multilingual counterparts trained on orders of magnitude more data. For a language like Arabic — with over 400 million native speakers, significant dialectal diversity, and historically underserved by commercial speech technology — this finding is consequential. The open-source release lowers the barrier for research groups across the Arab world to build on state-of-the-art foundations, and the diacritization findings open an important new application domain in Arabic literacy and educational technology.

B. Technical Architecture & Methodology

Model Architecture

ArTST is built upon SpeechT5 (Ao et al., 2022), a unified encoder-decoder Transformer originally developed for English. The key architectural innovation of SpeechT5 is its shared backbone design: a single encoder-decoder pair can accept both speech and text as input and produce both modalities as output, enabling pre-training and fine-tuning across diverse speech-text tasks without architectural modification. The ArTST adaptation consists of the following components:

- Shared Encoder: A multi-layer Transformer encoder with multi-head self-attention, positional embeddings, and feed-forward sub-layers. The encoder processes continuous speech frames from the Speech Pre-Net and learns deeply contextualized acoustic representations. During ASR fine-tuning, the encoder output is directly consumed by the decoder.
- Shared Decoder: A standard autoregressive Transformer decoder with cross-attention over encoder outputs. During ASR, the decoder generates Arabic text tokens one at a time, conditioned on the encoder's acoustic representations and previously generated tokens.
- Speech Pre-Net: A stack of 1D convolutional layers with GELU activations applied directly to raw waveforms, transforming the time-domain signal into continuous frame-level feature sequences at approximately 50 Hz (20ms per frame). This component replaces hand-crafted feature extraction pipelines.
- Speech Post-Net: A convolutional refinement network applied to the output mel-spectrogram during TTS generation; this component is inactive during ASR.
- Text Pre/Post-Nets: Learned embedding lookup tables for tokenized text, used as input to the decoder during TTS generation or as output targets during ASR fine-tuning.

For the ASR configuration specifically: raw waveform → Speech Pre-Net → Transformer Encoder → Transformer Decoder → Arabic text tokens. A CTC (Connectionist Temporal Classification) auxiliary

loss is applied at the encoder output in addition to the primary cross-entropy loss at the decoder output, following joint CTC-attention training as established by Watanabe et al. (2017).

Feature Extraction Method

ArTST operates directly on raw waveforms at 16 kHz sampling rate, making it an end-to-end model without a separate feature engineering stage. The Speech Pre-Net's convolutional layers learn to extract acoustic features from the raw signal, theoretically capturing fine-grained information that fixed filterbank representations (such as 80-dimensional log-Mel filter banks used by Whisper) may compress or discard. This approach has been validated in wav2vec 2.0 and HuBERT, and is adopted by ArTST following the same design philosophy.

Acoustic Modeling Approach

The acoustic model is the Transformer encoder. Pre-training follows a masked speech prediction strategy inspired by HuBERT: a proportion of speech encoder input frames are randomly masked, and the model is trained to reconstruct discrete speech units generated offline via k-means clustering of intermediate HuBERT feature representations. This self-supervised pre-training forces the encoder to develop rich, context-aware acoustic representations of Arabic phonetics and prosody before any supervised fine-tuning occurs.

Language Modeling Integration

The ArTST ASR results are reported without any external language model fusion (the paper explicitly states: "Results are shown without LM fusion"). The decoder does implicitly model short-range language context through its autoregressive generation mechanism and cross-attention, functioning as an implicit weak language model. The absence of external LM fusion is a deliberate methodological choice to isolate and evaluate the acoustic model's intrinsic capabilities. The authors note that integration with an external n-gram or neural Arabic language model is expected to further reduce WER and is left as future work.

Training Methodology

- Pre-training: Unsupervised masked speech prediction on MGB-2 1K (~1,000 hours) without requiring text transcriptions. Uses the Adam optimizer with a linear warmup phase followed by polynomial learning rate decay. Gradient clipping is applied for training stability.
- ASR Fine-tuning: Joint CTC + cross-entropy supervised training on MGB-2-1K paired speech-text data. The CTC loss is applied at the encoder output for fast convergence and as a regularization mechanism, while the cross-entropy loss at the decoder drives the final transcription quality.
- Dialect Identification Fine-tuning: A classification head is appended to the mean-pooled encoder output and trained with cross-entropy on the ADI17 dataset (17-class dialect classification task).
- Frameworks: The original implementation uses Fairseq (Facebook AI Research). Post-publication, the model was integrated into HuggingFace Transformers for broader accessibility, enabling fine-tuning via standard HuggingFace training pipelines.
- Hardware: Training conducted on NVIDIA A100 GPUs at MBZUAI (Mohamed bin Zayed University of Artificial Intelligence, Abu Dhabi).

Special Techniques for Arabic

- Diacritization study: The authors conduct dedicated experiments training the ASR model on diacritized transcriptions (with short vowel markers included in the output vocabulary). They demonstrate that ASR-based diacritization substantially outperforms standard text-based

post-processing diacritization models. This is a novel use of ASR as a linguistic normalization tool specific to Arabic.

- Buckwalter transliteration: For TTS experiments, Arabic script is converted to Buckwalter ASCII transliteration to avoid encoding issues in the SpeechT5 text processing pipeline. For ASR, standard Unicode Arabic script is used directly.
- Character/subword tokenization: The model uses a character-level or byte-pair encoding (BPE) vocabulary to handle Arabic's morphological complexity, avoiding the sparsity problems of word-level vocabularies in Arabic.

C. Dataset and Experimental Design

Datasets Used

Property	MGB-2 (Multi-Genre Broadcast 2)
Full Name	Multi-Genre Broadcast Challenge 2
Content	Arabic broadcast TV recordings from Al-Jazeera Arabic (news, sports, talk shows, documentaries, drama)
Total Size	~1,200 hours; ArTST uses the official 1,000-hour subset (MGB-2-1K)
Arabic Variety	~70% MSA, ~30% dialectal (Egyptian, Gulf, Levantine, North African)
Number of Speakers	Hundreds of professional broadcast speakers
Audio Format	16 kHz, single channel
Published By	Ali et al., IEEE SLT Workshop, 2016
Role in ArTST	Pre-training + ASR fine-tuning + ASR evaluation (train/dev/test splits)

Property	QASR (Qatari Arabic Speech Recognition Corpus)
Full Name	Qatari Arabic Speech Recognition corpus
Content	Al-Jazeera broadcast speech with automatic alignment using the Montreal Forced Aligner
Total Size	~2,000 hours
Arabic Variety	Primarily MSA with some Gulf dialectal content
Role in ArTST	Zero-shot cross-domain generalization evaluation (model not trained on QASR)

Property	CIArTTS (Classical Arabic TTS Corpus)
Content	Professionally recorded Arabic speech for TTS training
Size	~12 hours
Role in ArTST	TTS fine-tuning only (not used for ASR evaluation)

Property	ADI17
Full Name	Arabic Dialect Identification dataset covering 17 countries
Content	Short spoken utterances collected from YouTube across 17 Arabic-speaking nations
Role in ArTST	Dialect identification fine-tuning and evaluation

Data Preprocessing and Augmentation

- Audio normalization: All recordings resampled to 16 kHz mono. Amplitude normalization applied to handle volume variability across broadcast recordings.
- Speed perturbation: Three-way speed perturbation (0.9x, 1.0x, 1.1x speed factors) is applied during the pre-training phase, effectively tripling the apparent data diversity and improving robustness to natural speaking rate variation.
- SpecAugment: Time masking (random time segments zeroed out) and frequency masking (random frequency bands zeroed out) are applied to log-Mel spectrogram representations during ASR fine-tuning, following Park et al. (2019). This reduces overfitting and improves generalization to noisy test conditions.
- Text normalization: Arabic text is cleaned by removing non-Arabic characters, normalizing various Unicode forms of Arabic letters (e.g., multiple hamza forms are standardized), removing punctuation, and optionally retaining or stripping diacritics depending on the experimental configuration.

Train / Validation / Test Split

The paper adheres strictly to the predefined official MGB-2 splits to ensure comparability with previously published results. The training set is the MGB-2-1K subset (~1,000 hours), the development set is the standard MGB-2 development partition, and the test set follows the official MGB-2 benchmark evaluation protocol. For QASR, only the official test set is used for zero-shot evaluation, with no QASR data used during training.

Evaluation Metrics

- Word Error Rate (WER): Primary ASR metric, computed as (Substitutions + Deletions + Insertions) / Total Reference Words × 100. WER is the standard benchmark metric enabling cross-paper comparison.
- Character Error Rate (CER): Secondary metric particularly important for Arabic ASR, as CER is less sensitive to word segmentation inconsistencies (Arabic word boundaries are

sometimes ambiguous due to cliticization) and provides a finer-grained view of recognition accuracy.

- Dialect Identification Accuracy: Standard multi-class classification accuracy for the ADI17 17-dialect benchmark.
- Mean Opinion Score (MOS): Perceptual listening test score for TTS evaluation (irrelevant to the ASR analysis in this report).

D. Results and Performance Analysis

Primary ASR Results on MGB-2 Test Set

Model	WER (%)	Notes / Parameters
ArTST v1 (Proposed)	13.42	No LM fusion; monolingual Arabic pre-training on 1,000h
Whisper Medium (~300M params)	~28	Zero-shot; pre-trained on 680,000h multilingual
Whisper Large (~1.5B params)	~26	Zero-shot; pre-trained on 680,000h multilingual
MMS Large (Meta Massively Multilingual)	~40	Zero-shot multilingual

ArTST achieves 13.42% WER, which is on par with or exceeding the prior state-of-the-art on MGB-2, without using any external language model. Most strikingly, this result is achieved using approximately 1,000 hours of Arabic training data while Whisper large uses 680,000 hours of multilingual data. ArTST's WER is roughly half of Whisper large's zero-shot WER, providing compelling evidence for the monolingual pre-training hypothesis.

Dialect Identification Results on ADI17

ArTST achieves 94% accuracy on the ADI17 17-dialect classification benchmark, establishing a new state-of-the-art. This result is notable because the model was pre-trained only on MSA broadcast data, yet its internal representations are highly discriminative across 17 Arabic dialects — suggesting that the encoder captures deep phonological features that generalize beyond the MSA training domain.

Cross-Domain Generalization (QASR)

When evaluated zero-shot on QASR test utterances, ArTST produces qualitatively accurate transcriptions of MSA content, suggesting reasonable cross-domain acoustic generalization. The paper provides qualitative examples showing that ArTST correctly transcribes novel MSA utterances not present in the MGB-2 training distribution, including an example where ArTST produces the correctly diacritized form while the baseline SpeechT5 model produces an incorrect unrelated transcription.

Computational Efficiency

- ArTST's parameter count is substantially smaller than Whisper large (~1.5B params) and comparable to Whisper medium (~300M), yet achieves significantly better Arabic ASR performance — demonstrating strong parameter efficiency for Arabic-specific tasks.
- Pre-training requires a fraction of the compute budget of multilingual models: approximately 1,000 GPU-hours estimated for ArTST pre-training vs. the orders-of-magnitude larger budgets for Whisper and MMS.
- Inference: The autoregressive decoder introduces latency compared to CTC-only models, but this is acceptable for most Arabic speech applications.

E. Bias and Fairness Analysis

Dialect Bias

ArTST v1 exhibits a pronounced MSA bias. The model is pre-trained and fine-tuned exclusively on MGB-2, which is approximately 70% MSA broadcast speech. While the ADI17 dialect identification experiment demonstrates that the encoder is sensitive to dialectal differences, no ASR evaluation is performed on purely dialectal speech (e.g., Egyptian colloquial, Gulf dialectal, Moroccan Darija). The 12.8% WER figure applies specifically to the MGB-2 test set, which itself contains approximately 30% dialectal content — meaning performance on purely dialectal inputs is likely substantially worse. The authors explicitly acknowledge this limitation and announce plans for dialectal extensions (realized as ArTST v2 in October 2024).

Gender Bias

The paper does not provide any gender-stratified performance analysis. MGB-2 broadcast data reflects the demographics of Al-Jazeera Arabic programming, where the majority of on-air speakers are adult male presenters and correspondents. Female speakers are underrepresented in training data, making it highly probable that the model underperforms on female speech — a fairness concern shared by most broadcast-trained Arabic ASR systems. This is a significant methodological gap in the paper's evaluation.

Age Bias

No age-stratified analysis is presented. Broadcast TV data is dominated by professional adult speakers in their prime speaking years (approximately 25–55 years old). Child speech, elderly speech, and speech from non-native Arabic speakers are entirely absent from the training distribution. Performance on these underrepresented populations is expected to be significantly worse, yet is neither evaluated nor acknowledged in the paper.

Socioeconomic and Educational Factors

MGB-2 consists exclusively of formal broadcast speech from highly educated professional media presenters, capturing prestige MSA speech patterns with minimal hesitations, strong articulation, and standard pronunciation. Speakers from rural areas, speakers with non-standard regional accents, or speakers with limited formal education who use non-prestige speech forms are entirely absent from training. This socioeconomic bias means the model is optimized for formal speech and would likely struggle in practical deployment scenarios involving informal conversational Arabic.

Code-Switching Bias

ArTST v1 does not support Arabic-English or Arabic-French code-switching. The output vocabulary is Arabic-only; foreign words in code-switched utterances will be either omitted, misrecognized as phonetically similar Arabic words, or produce decoding errors. This is a critical limitation for Arabic speech processing in North Africa (Arabic-French mixing) and across the Arab world's educated urban populations (Arabic-English mixing). The authors acknowledge this and plan to address it in future versions.

Dataset Representation

MGB-2 represents only one broadcast media source (Al-Jazeera Arabic), one geographic production center, and one primary domain (broadcast TV journalism). The demographic, geographic, and topical diversity of the Arabic-speaking world — spanning 22 countries, dozens of dialects, and a vast range of social registers — is captured only minimally. As a result, ArTST's strong benchmark performance may not translate to deployments involving conversational Arabic, religious speech, customer service interactions, or medical transcription.

Authors' Acknowledgment of Fairness Concerns

The authors acknowledge the MSA-centric limitation and dialect bias explicitly, noting plans to develop dialectal Arabic and code-switched versions in future work. However, gender bias, age bias, and socioeconomic bias are not addressed or even mentioned in the paper, representing a gap in the fairness analysis relative to current best practices in responsible AI.

F. Strengths (Pros)

- State-of-the-art ASR performance on MGB-2 (13.42% WER) achieved with a compact model using only 1,000 hours of Arabic training data, outperforming multilingual models with 680x more training data.
- Fully open-source: pre-trained model, fine-tuned ASR/TTS/dialect ID models, and training code are all publicly available on HuggingFace and GitHub, enabling broad reproducibility and community building.
- Unified architecture supporting ASR, TTS, and dialect identification in a single pre-trained backbone, making the model versatile and reducing the total research overhead for Arabic speech systems.
- Novel contribution to diacritization research: demonstrating that ASR-based diacritization outperforms text-based post-processing methods, opening a new research direction with strong practical relevance.
- Awarded Best Paper at ArabicNLP 2023, providing strong peer validation of the work's quality and significance.
- Strong generalization to QASR and ADI17 demonstrates that the learned representations transfer beyond the MGB-2 training domain.
- Provides a rigorous and reproducible baseline for future Arabic ASR research, with clearly documented training procedures, evaluation splits, and model releases.

G. Limitations and Weaknesses (Cons)

- MSA-only focus: ArTST v1 is exclusively trained on MSA data and lacks demonstrated performance on dialectal Arabic, which constitutes the vast majority of everyday Arabic speech. This severely limits deployment in real-world conversational Arabic applications.
- No language model fusion: All results are reported without LM integration. Adding a strong Arabic language model would likely significantly reduce WER and is standard practice in competitive ASR systems; the absence makes fair comparison with LM-integrated baselines difficult.
- No gender, age, or socioeconomic fairness analysis: The paper fails to evaluate performance across demographic subgroups, leaving important fairness questions unanswered.
- Single data source dependency: Exclusive reliance on MGB-2 (Al-Jazeera Arabic) creates domain bias and limits robustness testing. Cross-domain evaluation on conversational Arabic, religious speech, or spontaneous speech is entirely absent.
- No code-switching support: The model cannot handle the prevalent Arabic-English or Arabic-French mixing common in educated Arabic speech across the region.
- Autoregressive decoding latency: The encoder-decoder architecture is inherently slower than CTC-only models at inference time, which may be a limitation for real-time Arabic ASR deployment.
- Limited baseline comparison: The paper compares primarily against Whisper and MMS in zero-shot conditions; a comparison against fine-tuned Whisper or wav2vec 2.0 XLS-R models would provide more informative context about the performance ceiling for Arabic ASR.

Paper 2: N-Shot Benchmarking of Whisper on Diverse Arabic ASR

Full Citation (IEEE): B. Talafha, A. Waheed, and M. Abdul-Mageed, "N-Shot Benchmarking of Whisper on Diverse Arabic Speech Recognition," in *Proc. INTERSPEECH 2023,* Dublin, Ireland, Aug. 2023, pp. 5092–5096. DOI: 10.21437/Interspeech.2023-1044

A. Summary and Context

Main Topic and Research Objectives

This paper conducts a comprehensive empirical benchmarking study of OpenAI Whisper, the state-of-the-art weakly supervised multilingual ASR model, across a diverse set of Arabic speech recognition conditions. In parallel, it evaluates XLS-R, a self-supervised cross-lingual speech model based on wav2vec 2.0 architecture, trained on 128 languages. The study covers the full spectrum of fine-tuning conditions — zero-shot (no Arabic task-specific data), few-shot (1–16 hours of Arabic fine-tuning data), and full fine-tuning — across all major publicly available Arabic ASR benchmarks. The overarching research objective is to provide the Arabic ASR community with a rigorous, reproducible characterization of how these modern foundation models behave under realistic Arabic deployment conditions, particularly in the context of Arabic's extreme dialectal diversity.

Specific Problem Addressed in ASR

The paper addresses three interconnected problems in Arabic ASR:

- Dialectal generalization: Arabic is not one language but a continuum of dialects (Egyptian, Gulf, Levantine, Moroccan, Algerian, Yemeni, and many others) that differ substantially in phonology, vocabulary, and morphology. The fundamental question is whether large multilingual models, pre-trained on predominantly MSA data, can generalize to dialectal Arabic without dialect-specific training data.
- Few-shot efficiency: Given the scarcity of labeled dialectal Arabic speech data, a critical practical question is how much labeled data is actually required to achieve acceptable performance on a new Arabic dialect. The paper systematically answers this by evaluating 1-hour, 2-hour, 4-hour, 8-hour, and 16-hour fine-tuning scenarios.
- Robustness to novel dialects: Five Arabic dialects (Algerian, Jordanian, Palestinian, Emirati, and Yemeni) are not publicly available in standard benchmarks. The paper collects original evaluation data for these dialects to test models on genuinely unseen conditions, providing a realistic assessment of generalization limits.

Key Contributions and Innovations

- First systematic n-shot evaluation (zero, few, full) of Whisper specifically for Arabic, covering all major publicly available Arabic speech datasets simultaneously in a single controlled study.
- Original evaluation data collection for five underrepresented Arabic dialects (Algeria, Jordan, Palestine, UAE, Yemen) — filling a significant gap in the public Arabic ASR evaluation landscape.
- Quantitative characterization of Whisper's dialectal sensitivity: demonstrating that zero-shot WER on Moroccan Arabic (MGB-5) is 82.97% — essentially random performance — while WER on the FLEURS dataset is 11.56%, revealing a dramatic 7:1 performance ratio that exposes the model's MSA/standard Arabic bias.

- Demonstration that XLS-R with default hyperparameters completely fails to converge on MGB-5 (Moroccan Arabic), revealing a critical robustness failure in self-supervised multilingual models for low-resource Arabic dialects.
- Practical finding that full fine-tuning consistently and substantially outperforms zero-shot and few-shot Whisper across all datasets, providing actionable guidance for Arabic ASR practitioners.

Significance and Potential Impact

The paper provides the Arabic ASR community with an essential benchmarking resource: a systematic, reproducible characterization of the most widely deployed ASR foundation models under realistic conditions. The quantitative exposure of Whisper's dialect bias — achieving 11.56% WER on clean FLEURS MSA but 82.97% WER on Moroccan Arabic — is a critical finding for responsible deployment of Arabic speech technology. The work has been widely cited as a reference benchmark for Arabic ASR evaluation methodology and directly motivates research into dialectal Arabic pre-training strategies.

B. Technical Architecture & Methodology

Model 1: Whisper (OpenAI, 2022)

Whisper is a large-scale weakly supervised encoder-decoder Transformer trained on approximately 680,000 hours of multilingual speech data collected from the internet, with approximately 17% non-English content. For Arabic specifically, Whisper's training data includes over 700 hours of Arabic speech for recognition and approximately 2,300 hours for translation, though the full composition of Arabic training data is not disclosed by OpenAI. The architecture consists of:

- Encoder: Multi-layer Transformer encoder processing log-Mel spectrogram features (80-dimensional filterbank) extracted from raw waveforms at 16 kHz. The encoder uses learned absolute positional embeddings.
- Decoder: Autoregressive Transformer decoder with cross-attention over encoder outputs. Decoding is guided by special task tokens specifying the language, task (transcribe vs. translate), and timestamp generation settings.
- Scale variants: Whisper is available in multiple sizes — Tiny (39M), Base (74M), Small (244M), Medium (307M), and Large (1.55B parameters). The paper evaluates Whisper Small (~244M) and Whisper Large (~1.55B).
- Training supervision: Weakly supervised — transcriptions are collected automatically from the internet without manual quality control, making Whisper's training data inherently noisy but extremely large-scale.

Model 2: XLS-R (Facebook AI Research, 2021)

XLS-R is a self-supervised cross-lingual speech representation model based on the wav2vec 2.0 architecture, scaled to 128 languages with a 1 billion parameter model trained on 436,000 hours of publicly available speech. The architecture consists of:

- Feature Extractor: Stacked 1D convolutional layers applied directly to raw waveforms, producing latent speech representations at 50 Hz.
- Context Network: Deep Transformer encoder with relative positional encodings, trained via a contrastive masked prediction objective (a portion of feature extractor outputs are masked,

and the context network must distinguish the true quantized representation of masked frames from distractors).

- CTC ASR Head: For downstream ASR fine-tuning, a linear CTC head is appended to the context network and trained with CTC loss on labeled speech data. The feature extraction layer is frozen during fine-tuning, while the Transformer encoder is updated.
- Language Model Decoding: The paper also evaluates XLS-R with an external statistical n-gram language model (XLS-R-LM), using KenLM for beam search decoding, demonstrating a consistent WER reduction of approximately 9 absolute WER points across datasets.

N-Shot Fine-Tuning Methodology

The paper defines three fine-tuning conditions that represent a progression of available labeled Arabic data:

- Zero-shot: Models are evaluated directly without any Arabic task-specific training. For Whisper, this uses the model's built-in multilingual decoding capability with language set to Arabic. For XLS-R, this is the base pre-trained model without any ASR fine-tuning.
- Few-shot: Whisper is fine-tuned on subsets of 1, 2, 4, 8, and 16 hours of labeled Arabic speech sampled from the training set of each target dataset.
- Full fine-tuning: Whisper is fine-tuned on the full available training set for each dataset (ranging from ~10 hours for MGB-3 to hundreds of hours for MGB-2).

Training Configuration

- Hardware: NVIDIA V100 GPUs. Due to Whisper large's memory requirements, the model is parallelized using DeepSpeed ZeRO Stage-2.
- Batch size: Single-device batch size of 32, effective batch size of 256 using gradient accumulation.
- Feature extraction layer frozen during fine-tuning for both Whisper and XLS-R, following standard practice.
- Training framework: HuggingFace Transformers and Trainer API, with DeepSpeed for large model parallelism.

Language Modeling Integration

XLS-R results are evaluated both without (XLS-R) and with (XLS-R-LM) an external statistical n-gram language model during CTC beam search decoding. The language model consistently reduces WER by approximately 9 absolute points, underscoring the importance of language model integration for Arabic CTC-based ASR. Whisper uses no external language model, relying on its decoder's implicit language modeling capacity.

C. Dataset and Experimental Design

Datasets Used

Dataset	Arabic Variety	Size (Training)	Purpose in Paper
Common Voice 6.1, 9.0, 11.0 (CV)	Mixed MSA/Dialectal	~15–21h train	Primary few-shot & full fine-tuning benchmark

MGB-2	~70% MSA, ~30% dialectal	~1,000h train	MSA broadcast ASR benchmark
MGB-3	Egyptian Arabic	~10.2h train	Dialectal (Egyptian) ASR benchmark
MGB-5	Moroccan Arabic (Darija)	~13h train	Dialectal (Moroccan) ASR; most challenging
FLEURS (Arabic)	MSA accented conditions	~10h train	Cross-lingual low-resource benchmark
AraYouTube (novel)	5 novel dialects: Algerian, Jordanian, Palestinian, Emirati, Yemeni	Evaluation only (no training data)	Robustness to unseen dialects

Data Preprocessing

- All audio resampled to 16 kHz mono. Duration trimming to maximum 30-second segments for Whisper compatibility (Whisper operates on fixed 30-second windows).
- Text normalization: removal of diacritics (harakat), normalization of Arabic letter variants, removal of non-Arabic characters and punctuation. Both WER and CER are reported after normalization.
- For Whisper fine-tuning: log-Mel spectrogram features (80 channels) are extracted by Whisper's built-in feature extractor. For XLS-R: raw waveforms are passed directly to the convolutional feature extractor.
- AraYouTube collection: five dialect-specific datasets are assembled by the authors from publicly available YouTube content for Algerian, Jordanian, Palestinian, Emirati, and Yemeni Arabic, each containing approximately 4–5 hours of transcribed speech.

Train / Validation / Test Splits

Official predefined splits are used for all standard benchmarks (MGB-2, MGB-3, MGB-5, FLEURS, Common Voice). For the novel AraYouTube dialect datasets, no training data exists — only evaluation sets are used to test zero-shot generalization. For few-shot experiments, training hours are randomly sampled from the full training partition of each dataset.

Evaluation Metrics

- Word Error Rate (WER): Primary metric, reported as WER / CER pairs in all result tables.
- Character Error Rate (CER): Secondary metric, particularly informative for Arabic where word segmentation can be ambiguous.
- Both metrics are computed after standard Arabic text normalization to ensure consistency across systems and datasets.

D. Results and Performance Analysis

Core Whisper Performance Results (WER% / CER%)

The following table reports Whisper Large's V2 performance across datasets and fine-tuning conditions:

Setting	CV11.0	MGB-2	MGB-5
Zero-shot	19.41 / 6.76	34.74 / 17.75	82.97 / 49.12
4h fine-tuning	15.47 / 4.72	24.43 / 12.54	56.24 / 23.41
Full fine-tuning	13.92 / 4.31	21.57 / 11.22	-
Whisper Large (zero-shot ref.)	13.28 / 4.23	15.49 / 8.62	53.82 / 22.99

Model	MGB-3 (Egyptian)	MGB-5 (Moroccan)	FLEURS
Whisper Small (zero-shot)	49.68 / 22.48	73.98 / 33.46	23.76 / 9.0
Whisper Large (zero-shot)	31.39 / 13.25	53.82 / 22.99	10.36 / 3.3
XLS-R (fine-tuned, no LM)	55.12 / 21.06	N/A (failed to converge)	N/A
XLS-R-LM (fine-tuned + n-gram LM)	48.41 / 19.6	N/A	N/A

Key Findings

- Whisper zero-shot performance on Moroccan Arabic (MGB-5) is catastrophically poor at 82.97% WER — effectively failing to recognize Moroccan Darija. This is the most striking bias result in the paper, revealing that Whisper's Arabic training data is overwhelmingly MSA/Egyptian, with Moroccan Arabic essentially absent.
- Whisper zero-shot outperforms fully fine-tuned XLS-R on all datasets, confirming the strength of Whisper's large-scale pre-training even without task-specific adaptation.
- Full fine-tuning consistently outperforms zero-shot and few-shot for Whisper across all datasets, confirming that dialect-specific labeled data substantially improves performance when available.
- XLS-R completely fails to converge on MGB-5 with default hyperparameters, attributed by the authors to the extreme phonological distance between Moroccan Darija and the MSA-dominated XLS-R pre-training data.
- Incorporating an n-gram language model with XLS-R (XLS-R-LM) reduces WER by approximately 9 absolute points consistently, highlighting the critical role of Arabic language modeling in CTC-based systems.

- On the five novel AraYouTube dialects, Whisper's zero-shot performance deteriorates significantly for Algerian, Jordanian, Palestinian, Emirati, and Yemeni Arabic — dialects the authors hypothesize were underrepresented or absent from Whisper's training data.

Comparison with Human Performance

Citing a related benchmarking study (Hussain et al., 2023), the paper notes that human performance on MGB-2 achieves approximately 8.9% WER — meaning even the best fine-tuned systems in this study (21.57% WER on MGB-2) remain approximately 12 absolute WER points behind human transcription accuracy on broadcast MSA Arabic. The gap widens substantially for dialectal Arabic.

E. Bias and Fairness Analysis

Dialect Bias — Central Focus of the Paper

Dialect bias is the central finding of this paper. The results demonstrate a dramatic performance gradient across Arabic varieties:

- FLEURS (MSA, clean conditions): 11.56% WER zero-shot — near-human-competitive performance.
- Common Voice 11.0 (mixed): 19.41% WER zero-shot — reasonable performance.
- MGB-2 (mixed MSA/dialectal broadcast): 34.74% WER zero-shot — substantial degradation.
- MGB-3 (Egyptian dialectal): 43.53% WER zero-shot — poor performance.
- MGB-5 (Moroccan Darija): 82.97% WER zero-shot — near-complete failure.

This 7:1 ratio between best-case (FLEURS) and worst-case (MGB-5) zero-shot performance directly quantifies the dialect bias embedded in Whisper's training data distribution. The paper makes this bias explicit and central to its contribution, unlike most prior Arabic ASR papers that evaluate only on MSA benchmarks.

Gender Bias

The paper does not analyze gender-based performance disparities. The multi-source datasets used (YouTube, broadcast TV, crowdsourced Common Voice) have varying and largely undocumented speaker gender distributions. No gender-stratified WER analysis is provided, which represents a missed fairness evaluation opportunity given the breadth of the study.

Age Bias

Age-stratified analysis is absent. The datasets range from professionally controlled recordings (FLEURS) to crowd-sourced contributions (Common Voice) with varying speaker age distributions. No age-based evaluation is performed.

Code-Switching

The paper does not evaluate Arabic-English or Arabic-French code-switching. All evaluation datasets are monolingual Arabic, despite the fact that code-switching is highly prevalent in educated urban Arabic speech across the Middle East and North Africa. This is a noted gap relative to the paper's goal of characterizing Whisper under realistic Arabic conditions.

Novel Dialect Coverage

The collection of AraYouTube evaluation data for five previously uncovered dialects (Algerian, Jordanian, Palestinian, Emirati, Yemeni) represents a significant positive contribution to fairness and coverage. These dialects represent tens of millions of speakers whose speech was not previously represented in publicly available Arabic ASR benchmarks. The paper explicitly acknowledges that Whisper's training data likely underrepresents these dialects, framing this as an equity concern.

Training Data Transparency

The paper notes a critical limitation in assessing Whisper's bias: OpenAI has not fully disclosed the composition of Whisper's Arabic training data (only that it includes CV9.0 and FLEURS). This opacity prevents systematic bias auditing and is acknowledged as a barrier to fairness analysis. The paper makes a methodological contribution by empirically characterizing the model's behavior in lieu of training data transparency.

Authors' Acknowledgment of Fairness Concerns

This paper is significantly stronger than ArTST in its acknowledgment of fairness concerns, particularly dialect bias. Dialect equity is the central motivation for the paper's design, the authors explicitly frame their novel dialect evaluation data as an equity contribution, and the results directly quantify performance disparities across Arabic varieties. Gender, age, and socioeconomic dimensions are not addressed.

F. Strengths (Pros)

- Comprehensive evaluation: All major publicly available Arabic ASR benchmarks are evaluated in a single controlled study, making this the most systematic Arabic ASR benchmarking work at the time of publication.
- Novel evaluation data: Collection of AraYouTube data for five underrepresented dialects fills a critical gap in the Arabic ASR evaluation landscape.
- Practical few-shot findings: Clear quantification of how much labeled data is needed to achieve reasonable performance at each dialect — directly actionable for practitioners building Arabic speech systems.
- Fairness-motivated design: Dialect bias is explicitly framed as the central concern, making the paper one of the first Arabic ASR papers to treat dialect equity as a primary evaluation criterion.
- Strong in INTERSPEECH 2023, a top-tier speech processing venue, ensuring methodological rigor through peer review.
- Both Whisper and XLS-R are freely available models, making all experiments fully reproducible.
- XLS-R convergence failure on MGB-5 is reported transparently rather than hidden, contributing valuable negative results to the field.

G. Limitations and Weaknesses (Cons)

- No monolingual Arabic pre-training baseline: The paper does not include Arabic-specific models like ArTST, wav2vec 2.0 fine-tuned on Arabic, or other monolingual systems. All

evaluated models are multilingual, limiting the paper's ability to assess the ceiling performance for Arabic ASR.

- No diacritization analysis: Unlike ArTST, this paper treats Arabic text as undiacritized throughout, not investigating the diacritization challenge that is linguistically fundamental to Arabic ASR.
- Limited AraYouTube dataset size: The five novel dialect evaluation sets are relatively small (~4–5 hours each) and collected from YouTube with automatic transcription quality. This introduces noise into the dialect robustness evaluation.
- No gender, age, or socioeconomic fairness analysis despite the paper's fairness framing — the fairness contribution is limited to dialect coverage.
- No code-switching evaluation despite prevalence in real-world Arabic speech.
- Whisper fine-tuning results are not compared against ArTST or other Arabic-specialized models, missing an opportunity to assess whether multilingual fine-tuning can match monolingual pre-training efficiency.
- MGB-5 full fine-tuning with Whisper achieves only 53.82% WER — still very poor performance on Moroccan Arabic — suggesting that even with the full 13 hours of available training data, Moroccan Arabic ASR remains fundamentally unsolved by current models.

3. Comparative Analysis

This section provides a structured comparative evaluation of the two papers across all dimensions specified in the project requirements. The two papers represent fundamentally different approaches to Arabic ASR: Paper 1 (ArTST) advocates for monolingual Arabic-specific pre-training from scratch, while Paper 2 (N-Shot Whisper Benchmarking) evaluates how large-scale multilingual models can be adapted to Arabic through fine-tuning. Understanding the trade-offs between these paradigms is central to Arabic speech processing research.

3.1 Architectural Differences

Dimension	ArTST vs. Whisper/XLS-R
Pre-training strategy	ArTST: Unsupervised masked speech prediction on Arabic-only data from scratch. Whisper: Weakly supervised on 680,000h multilingual web data. XLS-R: Self-supervised contrastive learning on 436,000h multilingual data.
Feature extraction	ArTST: Raw waveform → 1D CNN (learned). Whisper: Raw waveform → 80-dim log-Mel spectrogram (fixed). XLS-R: Raw waveform → 1D CNN (learned, similar to ArTST).
Decoder architecture	ArTST: Autoregressive Transformer decoder (encoder-decoder). Whisper: Autoregressive Transformer decoder (encoder-decoder). XLS-R: CTC linear head only (encoder-only), no decoder.
Language modeling	ArTST: Implicit (decoder). Whisper: Implicit (decoder). XLS-R: External n-gram LM via CTC beam search.
Multi-task capability	ArTST: Single backbone for ASR, TTS, and dialect ID. Whisper: ASR + speech translation. XLS-R: CTC-based ASR only.
Arabic specialization	ArTST: Fully Arabic-specific; trained only on Arabic data. Whisper/XLS-R: General multilingual; Arabic is a small fraction of training.

3.2 Dataset Differences

Dimension	ArTST (Paper 1)	Whisper Benchmark (Paper 2)
Training data	MGB-2 1K subset (1,000h MSA broadcast)	MGB-2, MGB-3, MGB-5, CV (multi-dialect; 10–1,000h)

Dialect coverage	MSA only (+ 30% incidental dialect in MGB-2)	MSA + Egyptian + Moroccan + multi-dialect CV + FLEURS
Novel dialects	None (MSA only)	5 novel dialects: Algerian, Jordanian, Palestinian, Emirati, Yemeni
Dataset diversity	Single source (Al-Jazeera Arabic)	Multiple sources: TV, crowdsourced, YouTube, academic
Total evaluation sets	MGB-2 test, QASR zero-shot, ADI17	6 standard datasets + 5 novel dialect sets = 11 total
Data size advantage	Small but highly curated Arabic corpus	Much broader coverage but lower per-dialect depth

3.3 Performance Comparison

Direct numerical comparison is possible on the shared MGB-2 benchmark:

Model	MGB-2 WER (%)	MGB-3 WER (%)	MGB-5 WER (%)
ArTST v1	12.8 (no LM)	Not evaluated	Not evaluated
Whisper Large (zero-shot)	34.74	43.53	82.97
Whisper Large (full fine-tune)	21.57	~53.82	53.82
XLS-R (fine-tuned + LM)	N/A (did not train on MGB-2 in Paper 2)	48.41	Failed to converge

On MGB-2, ArTST's 12.8% WER is substantially lower than Whisper Large's zero-shot 34.74% and even significantly better than Whisper Large's full fine-tuned 21.57% WER. This comparison is particularly striking because ArTST uses only MGB-2 training data while Whisper was exposed to 680,000 hours of multilingual speech during pre-training. However, ArTST's performance advantage is domain-specific: it is trained and evaluated on the same MGB-2 distribution, while Whisper's full fine-tuning in Paper 2 uses the same data. A fairer comparison would fine-tune Whisper on the same MGB-2 training partition used by ArTST, which Paper 2 does not do. Notwithstanding this caveat, ArTST's architectural design philosophy — monolingual pre-training from scratch — demonstrates a clear performance advantage on the Arabic MSA benchmark, even without language model integration.

For dialectal Arabic, ArTST has not been evaluated (v1), while Paper 2 comprehensively documents Whisper's substantial degradation on dialectal benchmarks. This represents a complementary finding: ArTST is state-of-the-art for MSA but untested on dialects; Whisper is comprehensively characterized on dialects but significantly inferior on MSA.

3.4 Computational Efficiency

Dimension	Comparison
Parameter count	ArTST: ~Comparable to Whisper small (~155M). Whisper Large: ~1.55B. XLS-R: ~1B. ArTST achieves SOTA performance at small-scale parameter count.
Pre-training compute	ArTST: Estimated ~1,000 GPU-hours on A100s for MGB-2 1K pre-training. Whisper: Undisclosed but estimated millions of GPU-hours for 680,000h. XLS-R: Extensive compute on 436,000h.
Training data required	ArTST: Only 1,000 hours of Arabic. Whisper: 680,000 hours multilingual. XLS-R: 436,000 hours multilingual. ArTST is dramatically more data-efficient for Arabic.
Fine-tuning efficiency	Paper 2 shows that 4 hours of fine-tuning data is sufficient for Whisper to achieve reasonable CV performance, while full dataset fine-tuning is needed for dialectal benchmarks.
Inference speed	Both ArTST and Whisper use autoregressive decoding — similar inference latency. XLS-R with CTC is faster (non-autoregressive) but requires external LM for good performance.
Resource accessibility	ArTST's smaller pre-training footprint makes it more accessible to research groups without large GPU clusters. Whisper is available for inference but not practical to pre-train from scratch.

3.5 Practical Deployment Considerations

From a real-world deployment perspective, the two approaches have distinct trade-offs:

- ArTST is more suitable for MSA-focused production applications (news transcription, formal document dictation, broadcast monitoring) where data is predominantly MSA. Its compact size, strong MGB-2 performance, and open-source availability on HuggingFace make it practically deployable today.
- Whisper (Paper 2's primary subject) offers greater out-of-the-box dialectal coverage for non-MSA Arabic when fine-tuning data is available, despite its worse zero-shot performance. For teams with access to labeled dialectal data, fine-tuned Whisper is a practical solution for multi-dialect Arabic ASR.

- XLS-R (evaluated in Paper 2) is less suitable for production Arabic deployment due to its failure to converge on Moroccan Arabic and its consistently lower performance compared to fine-tuned Whisper, even with language model integration.
- For low-resource dialectal deployment scenarios with minimal labeled data (1–4 hours), Paper 2 shows that Whisper's few-shot performance offers a practical starting point, but quality remains limited for distant dialects.

3.6 Robustness to Linguistic Challenges

Linguistic Challenge	ArTST (Paper 1)	Whisper/XLS-R (Paper 2)
Dialectal variation	Not evaluated; expected to degrade on non-MSA dialects	Comprehensively characterized; severe degradation on Moroccan, moderate on Egyptian
Diacritization	Actively investigated; ASR-based diacritization demonstrated to outperform text-based methods	Not addressed; all evaluations use undiacritized text
Morphological complexity	Addressed via character/subword tokenization; competitive WER suggests effective handling	Not specifically addressed; standard text normalization only
Code-switching	Not supported (Arabic-only vocabulary)	Not evaluated (all monolingual Arabic datasets)
Accented MSA	Not evaluated	Evaluated via FLEURS accented MSA conditions; performance maintained reasonably
Novel unseen dialects	Not evaluated	Comprehensively evaluated on 5 novel dialects; significant performance degradation confirmed

3.7 Bias and Fairness Comparison

Fairness Dimension	Comparison

Dialect bias acknowledgment	Paper 2 is significantly stronger: dialect bias is the central research question, quantified with 11 datasets. Paper 1 acknowledges MSA focus but does not quantify dialectal performance gaps.
Gender bias	Neither paper evaluates gender-stratified performance. This is a shared weakness.
Age bias	Neither paper evaluates age-stratified performance. This is a shared weakness.
Novel dialect coverage	Paper 2 uniquely contributes new evaluation data for 5 underrepresented Arabic dialects, making a direct equity contribution. Paper 1 does not address underrepresented dialects.
Socioeconomic bias	Neither paper addresses performance on informal, low-prestige speech registers or non-native speakers.
Training data transparency	ArTST: fully transparent (MGB-2 1K, publicly available). Paper 2: limited by OpenAI's non-disclosure of Whisper's full training composition.
Overall fairness score	Paper 2 is more thorough in fairness analysis due to its dialect-equity framing and novel data contribution. Paper 1's fairness analysis is more limited despite strong technical contributions.

3.8 Overall Assessment

The two papers are complementary rather than competing in their contributions to Arabic ASR research:

- ArTST (Paper 1) makes the stronger technical contribution for MSA Arabic ASR, establishing that monolingual Arabic-specific pre-training from scratch can outperform large multilingual models by a factor of 2x on MGB-2 while using a fraction of the parameters and training data. Its technical novelty (unified speech-text transformer for Arabic), strong performance (12.8% WER), and open-source accessibility make it the stronger paper for practitioners building MSA Arabic speech systems.
- The N-Shot Whisper Benchmarking paper (Paper 2) makes the stronger methodological and fairness contribution, providing the Arabic ASR community with a rigorous, reproducible characterization of the leading foundation models across the full spectrum of Arabic dialectal diversity. Its documentation of Whisper's catastrophic failure on Moroccan Arabic (82.97% WER zero-shot), its collection of novel dialect evaluation data, and its systematic few-shot analysis make it an essential reference for responsible Arabic speech technology deployment.

Taken together, these papers paint a complementary picture: monolingual pre-training (Paper 1) is the right approach for MSA excellence, while systematic dialectal benchmarking (Paper 2) reveals that neither multilingual nor monolingual approaches have solved the full Arabic dialectal ASR problem. The natural next step — building monolingual or bilingual pre-trained models for individual

Arabic dialects using the benchmarking methodology of Paper 2 — is precisely the direction taken by ArTST v2 (October 2024), which pre-trains on 17 Arabic dialects.

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